

BENJAMIN L. HANSON

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EDUCATION

University of California San Diego - La Jolla, CA

Current Ph.D., Expected May 2027

Major: Aerospace Engineering

Field: Dynamic Systems and Controls

University of California San Diego - La Jolla, CA

M.S., June 2024

Major: Aerospace Engineering

GPA: 3.744

Field: Dynamic Systems and Controls

Colorado School of Mines - Golden, CO

B.S., May 2022

Major: Engineering Physics

Summa Cum Laude, GPA: 3.93

Minor: Robotics and Intelligent Systems

Area of Special Interest: Space and Planetary Sciences Engineering

RELEVANT EXPERIENCE

NSTGRO Visiting Technologist Experience - Jet Propulsion Laboratory

June 2024/25 - August 2024/25

- Two, 10-week summer internships in Section 392K/324G: Mission Design and Navigation System Engineering
- **Summer 2025:** Discovered a strong relationship between the validity time of the Gaussian assumption and the unscented transform's skewness for low fidelity astrodynamics systems, which can be used for efficient-yet-accurate hybrid estimation
- **Summer 2024:** Applied non-Gaussian ensemble and grid-based techniques to the state uncertainty propagation of distant prograde orbits in the Saturn-Enceladus system as a proof of concept for outer planetary orbilander estimation and navigation; facilitated efficient communication between JPL's Monte library and C source code via Python wrapper

Air Force Office of Scientific Research (AFOSR) Summer Faculty Fellowship Program

July 2023 - September 2023

- Performed research at the Air Force Maui Optical and Supercomputing (AMOS) site hosted by the 15th Space Surveillance Squadron
- Developed a thorough validation of the landscape of numerical integrators and their respective accuracies relating to the state propagation of cislunar satellites in near-rectilinear halo orbits (NRHOs)
- Used publicly available ephemerides of CAPSTONE to detect orbital-maintenance maneuvers (OMMs) for numerical integration accuracy analysis, specifically comparing REBOUND, ASSIST, and GMAT

Auger@TA Ultra High Energy Cosmic Ray Detector Simulation Analysis

August 2021 - May 2022

- Collaboration between the Pierre Auger Observatory in Argentina and the Telescope Array in Utah to calibrate energy detection method variations
- Collected data using Linux Offline simulations with a varying number of photomultiplier tubes to determine if results were similar
- Visualized simulation data on the reliability of a single PMT detector vs. a triple PMT detector via Python
- Principal Investigator: Dr. Frederic Sarazin, Colorado School of Mines Physics Department

National Science Foundation REU - University of Florida, Astronomy Department

May 2021 - August 2021

- Developed and gave presentation on decomposing galactic spectral energy distributions and inferring properties from the simple stellar populations that make up the galaxies
- Utilized Bayesian statistics programming, specifically Markov Chain Monte Carlo to find spectrum best fit for age/metallicity
- Used HiPerGator, UF supercomputer, along with other Python simulation tools and techniques to model galactic behavior
- Principal Investigator: Dr. Paul Torrey, University of Florida Astronomy Department

UC San Diego MAE143B Linear Control Teaching Assistant

August 2025 - September 2025

- Duties included holding office hours, developing solutions sets to homeworks and exams, grading homework and exams, and presenting in lecture

HONORS AND AWARDS

- **2026 ASME Graduate Student Video Competition, 1st Place**, "[Predicting a Potential Lunar Impact: A Hybrid, Measurement-free Approach to Tracking Asteroid YR4](#)" (March 2026)
- **Student Travel Award for the 2026 SIAM Conference on Uncertainty Quantification**, "A Parallel Grid-based Algorithm for Orbital Uncertainty Propagation" (March 2026)
- **NASA Space Technology Graduate Research Opportunities (NSTGRO) Fellow** (Grant # 80NSSC23K1219), "A Grid-Based Bayesian Approach to Uncertainty Propagation for Icy-Moon Missions" proposal (August 2023 - August 2027)
- **Chambliss Astronomy Achievement Student Award Honorable Mention**, 240th AAS Meeting Presentation on "A Flexible Approach to Fitting Galactic Spectral Energy Distributions" (June 2022)
- **Colorado School of Mines Undergraduate Research Fellow (MURF)**, "Assembly of Microparticles for Robots and Composite Materials Under Combined Electric and Magnetic Fields" (August 2020 - May 2022)

- **Colorado School of Mines Physics Undergraduate Research Symposium: Best Individual Technical Research Project**, “Optimizing the Selection of Reconstructed Events in Auger@TA for Cross-Calibration Purpose Through Simulations” (April 2022)
- **Colorado School of Mines Dean’s List** (August 2018 - May 2022)

PUBLICATIONS

Refereed Journal Publications

- J6 **Hanson, B.L.**, Ely, T.A., Bewley, T.R., Rosengren, A.J.: Bayesian benchmarking of GBES applied to outer planet orbiter estimation. *Journal of Guidance, Control, and Dynamics* **49**(1), 240-246 (2026), <https://doi.org/10.2514/1.G009146>
- J5 **Hanson, B.L.**, Rubio, C., García-Gutiérrez, A., Bewley, T.: GBES-GPU: An efficient parallel GPU algorithm for high-dimensional nonlinear uncertainty propagation. *Computer Physics Communications* **317**, 12 (2025), <https://doi.org/10.1016/j.cpc.2025.109819>
- J4 **Hanson, B.L.**, Zhao, M., Bewley, T.R.: An extensible framework for the probabilistic search of stochastically-moving targets characterized by generalized Gaussian distributions or experimentally-defined regions of interest. *Communications in Statistics - Theory and Methods* **54**(17), 5480-5505 (2025), <https://doi.org/10.1080/03610926.2024.2439999>
- J3 Haque, M.A., Maestas, J.R., Zhu, X., **Hanson, B.L.**, Wu, D.T., Wu, N.: High-density and well-aligned hierarchical structures of colloids assembled under orthogonal magnetic and electric fields. *ACS Nano* (2025), <https://doi.org/10.1021/acsnano.4c11957>
- J2 Floriano, B.R., **Hanson, B.L.**, Bewley, T., Ishihara, J.Y., Ferreira, H.C.: A novel policy for coordinating a hurricane monitoring system using a swarm of buoyancy-controlled balloons trading off communication and coverage. *Engineering Applications of Artificial Intelligence* **139**, 109495 (2025), <https://doi.org/10.1016/j.engappai.2024.109495>
- J1 Zhu, X., Gao, Y., Mhanna, R., Yang, T., **Hanson, B.L.**, Yang, X., Gong, J., Wu, N.: Synthesis and propulsion of magnetic dimers under orthogonally applied electric and magnetic fields. *Langmuir* **37**, 9151-9161 (2021), <https://doi.org/10.1021/acs.langmuir.1c01329>

Submitted Journal Publications

- S1 **Hanson, B.L.**, Carton, L., Bewley, T.R., Ely, T.A., Rosengren, A.J.: Hybrid, ephemeris-quality, measurement-free estimation of the potential 2024 YR4 lunar impact. Submitted to the *Journal of Astronautical Sciences* (2026), <https://doi.org/10.21203/rs.3.rs-8196369/v1>

Conference Publications

- C2 **Hanson, B.L.**, Rosengren, A.J., Bewley, T.R., Ely, T.A.: Non-Gaussian recursive Bayesian filtering for outer planetary orbilander navigation. In: *AAS/AIAA Space Flight Mechanics Meeting*, p. 194 (2025), <https://doi.org/10.13140/RG.2.2.33100.73603>
- C1 **Hanson, B.L.**, Rosengren, A.J., Bewley, T.R.: State estimation of chaotic trajectories: A higher-dimensional, grid-based, Bayesian approach to uncertainty propagation. In: *AIAA SCITECH 2024 Forum*, p. 0426 (2024), <https://doi.org/10.2514/6.2024-0426>

ORAL PRESENTATIONS

- P9 *A Parallel Grid-based Algorithm for Orbital Uncertainty Propagation*, SIAM UQ26: Uncertainty Quantification in Astrodynamics Minisymposium, Minneapolis, Minnesota (March 2026)
- P8 *Efficient Prediction of the Gaussianity Validity Time in the Circular Restricted Three-Body Problem*, COPERNAICUS Seminar, On-line (September 2025)
- P7 *GBES-GPU: An efficient parallel GPU algorithm for high-dimensional nonlinear uncertainty propagation*, Cassyni Computer Physics Seminar Series, <https://doi.org/10.52843/cassyni.m84f85> (September 2025)
- P6 *Predicting the Temporal Limits of Gaussianity in the Saturn-Enceladus System with the Unscented Transform*, Jet Propulsion Laboratory, Pasadena, CA (July 2025)
- P5 *Non-Gaussian Recursive Bayesian Filtering for Outer Planetary Orbilander Navigation*, Jet Propulsion Laboratory, Pasadena, CA (August 2024)
- P4 *xGEO Numerical Integrator Analysis and Maneuver Detection*, REBOUND Conference 2024, Online (July 2024)
- P3 *On the Validity of the Gaussian Assumption in the Jovian System: Evaluating Linear and Nonlinear Filters for Measurement-sparse Estimation*, 6th International Workshop on Key Topics in Orbit Propagation Applied to SSA, Université d'Artois, Arras, France (June 2024)
- P2 *A Flexible Approach to Fitting Galactic Spectral Energy Distributions*, 240th AAS Meeting, Pasadena, CA (June 2022)
- P1 *Synthesis and Assembly of Anisotropic Particles*, Colorado School of Mines Undergraduate Research Symposium, Golden, CO (April 2022)